GATEWAY's Complete Guide to Local Digital Education

"Dear Educator and Student, you are holding the **key** to digital education. This USB is the ***gateway*** to computer science, containing everything you need to teach and learn the subject offline. "

Package Contents:

1. USB with main website that can run offline.
2. Dashboard to track progress (on website).
3. Hundreds of learning resources - Courses, videos, textbooks
4. Practice Arena - coding challenges with immediate feedback.
5. Installation wizard to help set up any computer.
6. Multilingual support, e.g. English, isiZulu, isiXhosa, Sesotho.
7. Teacher & Student training program.
8. Lesson Plan

Quick Start Up Guide

5-Minute Setup:

- Step 1: System Check

1. Computer with Windows 7+ or Ubuntu Linux or Android Phone
2. At least 2GB free space
3. Modern web browser (Chrome/Firefox/Edge & mobile versions)

- Step 2: Installation

1. Insert USB drive into computer.
2. Open USB folder called "Gateway Offline."
3. Double-click "START HERE"
4. Wait for green "Ready" message (about 1 minute)

- Step 3: Access Platform

1. Open web browser
2. Type: localhost:8080 in address bar
3. Welcome to your offline learning hub!

- Step 4: First Login
    1. Teacher username: admin
    2. Teacher password: teach2025
    3. Change password after first login

<u>Classroom management & teacher dashboard</u>

Creating Student Accounts:

1. Go to "Manage Students" tab

2. Click "Add New Student"

3. Enter: First Name, Last Name, Preferred Language

4. System generates unique student ID

5. Print student login cards or write them down

Student Login Process:

- Username: student[number] (e.g., student01)
- Password: learn2025
- Students change passwords after first login

Tracking Progress Offline

View Student Progress:

- Progress Dashboard: See completion rates for all students
- Skill Mastery: Track which concepts students have mastered
- Time Tracking: Monitor how long students spend on each topic
- Problem Areas: Identify topics where multiple students struggle

Generating Reports:

1. Select students or entire class.
2. Choose report type: Progress Report, Skill Gap Analysis, Completion Certificate
3. Click "Generate Report" - creates PDF for printing or sharing

Assignment Management

Creating Offline Assignments:

1. Go to "Assignments" tab
2. Click "New Assignment"
3. Select from 500+ pre-loaded exercises
4. Set due dates and instructions
5. Assign to specific students or entire class

Grading Student Work:

- Auto-grading for coding exercises
- Manual feedback system for projects
- Progress tracking across assignments

Curriculum Navigation

Learning Paths Available:

1. Absolute Beginner (0-3 months)

- Computer basics and digital literacy
- Introduction to computational thinking
- First steps in Python programming
- Building simple websites with HTML

2. Intermediate (3-9 months)

- Python programming mastery
- Web development (HTML, CSS, JavaScript)
- Database fundamentals
- Problem-solving algorithms

3. Advanced (9+ months)

- Data science and analysis
- Mobile app development
- Software engineering principles
- Project development

<u>Resource Types</u>

Interactive Courses:

- Self-paced with built-in exercises
- Video lessons (downloadable)
- Progress saving automatically.
- Multilingual subtitles available

Practice Arena:

- 300+ coding challenges
- Instant feedback and hints
- Multiple difficulty levels
- Progress tracking

Digital Library:

- 50 computer science textbooks
- University lecture notes
- Programming references
- Project idea catalogues

<u>Classroom Implementation</u>

Computer Lab Setup:

- Install platform on each computer using USB duplication
- Create student accounts in batches
- Set up progress tracking for entire class
- Designate a "sync master" computer for aggregating progress

Single Computer Setup:

- Use projector for group lessons
- Rotate students for hands-on practice
- Create group projects and collaborations
- Use printed exercises for offline practice

Teaching Without Internet:

Daily teaching structure

8:00-8:30 → Group instruction (projector)

8:30-9:30 → Hands-on practice (individual computers)

9:30-10:00 → Group discussion and problem-solving

10:00-10:30 → Progress review and next steps

Weekly Planning:

- Monday: introduction of a new concept
- Tuesday: practice exercises
- Wednesday: do a project
- Thursday: reviewing and assessment
- Friday: creative applications

Assessment Strategies:

Formative Assessments

- Daily practice exercises in Practice Arena
- Weekly coding challenges
- Peer code reviews
- Progress report reviews

Summative Assessments:

- Project submissions on monthly basis
- Skill mastery demonstrations
- Teaching concepts to peers

Synchronizing & backing up data

Automatic Progress Saving:

- progress saves locally every few minutes
- teachers can force save from dashboard
- backup files created daily

Transferring Progress:

1. exporting student data to USB drive
2. importing data on another computer
3. merging progress from multiple sources

Regular Maintenance:

- backing up student progress to a secondary USB
- clearing temporary files
- updating local database if new content is available
- full backup on a monthly basis

Multilingual Support

For teachers:

1. Go to Settings → Language Preferences
2. Select interface language
3. Choose default language for new students
4. Save preferences

For Students:

1. Students can select language preferences during their first login
2. Can switch languages anytime
3. Content available in multiple languages
4. Programming concepts taught with local language explanations

Supported Languages:

- ➔ English, isiZulu, isiXhosa, Sesotho, and Afrikaans
- ➔ Can expand to other languages in time.

Recognition System:

- Award Digital Badges for completed modules
- Progress certificates
- Skill mastery awards
- Project Showcase recognition
- Peer Recognition programs

Graduation Pathways:

- Digital Literacy Certificate - 3 months
- Programming Fundamentals Certificate - 6 months
- Web Developer Certificate - 9 months
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"We are not just teaching computer skills - we are opening doors to futures, building confidence, and creating the next generation of African women technology leaders."